Lab 02: Evaluate a Detection Rule

Practice cases, confusion-matrix metrics, and transparent reporting

Objective

Evaluate a simple rule against labelled training scenarios. The important result is not a high score by itself; it is understanding what the rule catches, what it misses, and which labels or assumptions affect the result.

USE ONLY TRAINING DATA The bundled cases are synthetic and intended for learning. They are not measured real-world attack performance.

Step 1 — Review the included cases

Open website/public/labs/detection-evaluation-cases.csv, or download the CSV from this lab card. Each row includes a case ID, a labelled scenario, process, command line, and rationale. The companion Python script applies a deliberately narrow sample rule and calculates its results. All cases are synthetic.

From the website folder, run: python public/labs/evaluate-detection-cases.py public/labs/detection-evaluation-cases.csv

Step 2 — Count outcomes

Actual label	Rule alerts	Classification
Suspicious	Yes	True positive (TP)
Benign	Yes	False positive (FP)
Benign	No	True negative (TN)
Suspicious	No	False negative (FN)

Step 3 — Use the metrics

Metric	Formula	What it tells you
Accuracy	(TP + TN) / all cases	Overall correctness; can mislead on imbalanced data.
Precision	TP / (TP + FP)	How many alerts were true positives in the labelled set.
Recall	TP / (TP + FN)	How many labelled suspicious cases were detected.
Specificity	TN / (TN + FP)	How many labelled benign cases were correctly ignored.
F1	2 × P × R / (P + R)	Harmonic mean of precision and recall.

Step 4 — Experiment

1. Record the baseline counts (TP, FP, TN, FN) before changing a rule.
2. Inspect each false positive. Is the rule too broad, or is context missing?
3. Inspect each false negative. What relevant field or event pattern is missing?
4. Change one condition at a time in your own local practice copy.
5. Re-run the same dataset and compare counts. Keep both baseline and revised results.
6. Do not tune on a small set and then present that same score as independent validation.

Step 5 — Write a reproducible report

- Dataset filename and version.
- Number of cases and label distribution.
- Rule logic and required log fields.
- TP, FP, TN, FN and calculated metrics.
- Two examples of false positives or false negatives, if present.
- Limitations, environment, and the next experiment.

ADVANCED HABIT Separate development data from a held-out evaluation set when the dataset is large enough. Review labels, duplicates, and scenario coverage.